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TOPOLOGICAL DATA ANALYSIS APPROACH ON FLOOD DETECTION IN ALOR SETAR, KEDAH

PRESENTED BY:

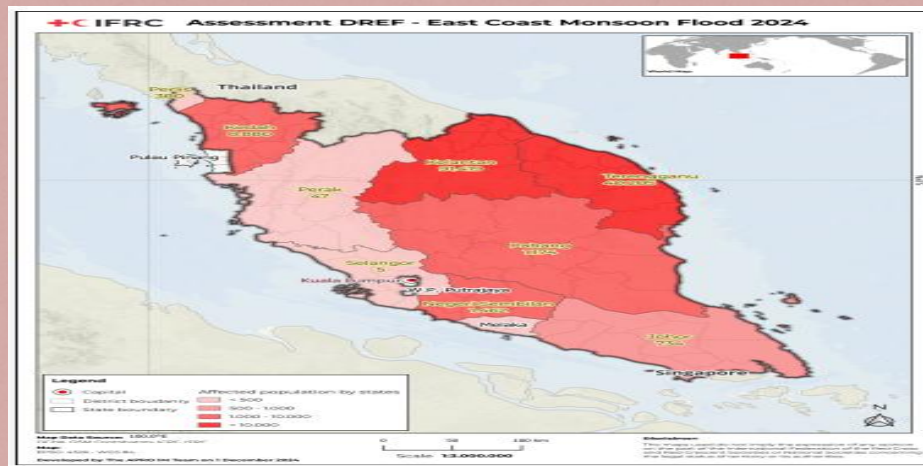
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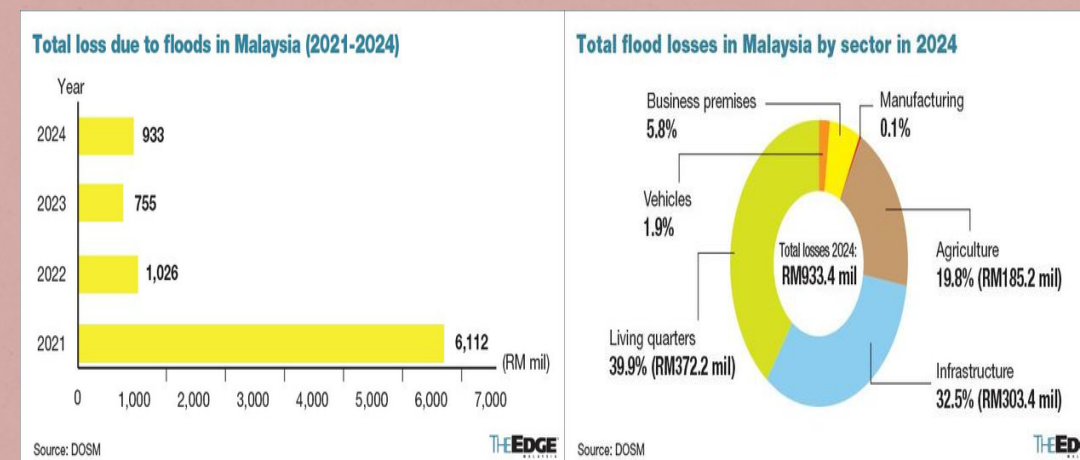
FLOOD DETECTION IN KEDAH

Floods occur when water overflows onto normally dry land, caused by heavy rain or human activities like deforestation and urbanization.



Monsoon season usually occur from November till March.
Affected State : Kelantan, Pahang, Selangor, Kedah

Floods have led to considerable losses in various sectors, particularly in agriculture (19.8%) and infrastructure (32.5%) in Malaysia, as indicated by the statistics from 2021-2024.

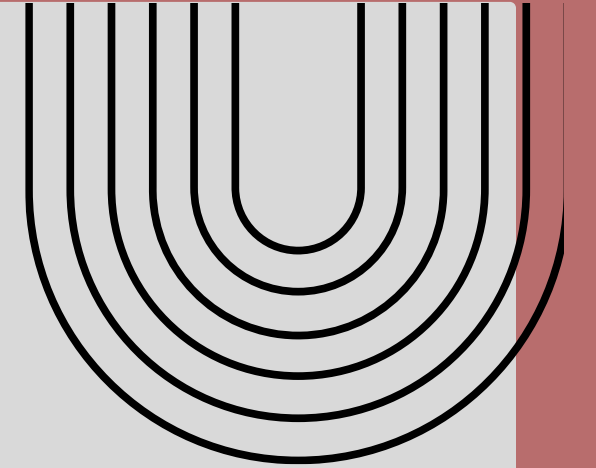


(Source : Department of Statistics Malaysia)



The federal government has allocated significant funding (RM79.5 million) to flood mitigation projects like the Sungai Kedah / Sungai Anak Bukit RTB Project, expected to be ready by September 2024.

PROBLEM STATEMENTS



Existing flood prediction models rely on data like wind speed and rainfall volume.



No flood prediction model using such mathematical methods has been implemented in Kedah to date.



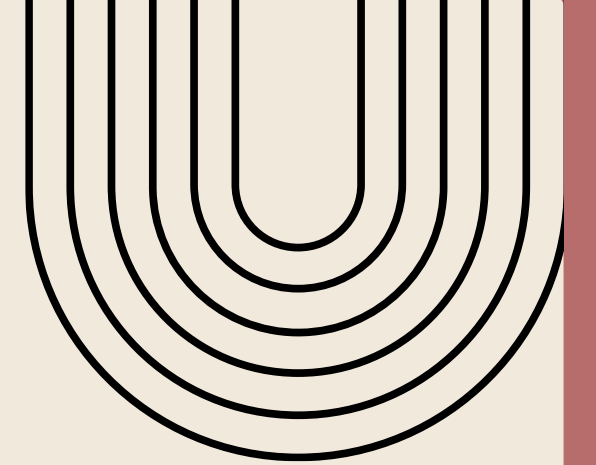
The accuracy of the proposed flood prediction model for Kedah needs to be analyzed and compared against existing benchmark studies.





OBJECTIVE

- 1 To identify the significant features of flood using TDA
 - 2 To enhance flood forecasting and early warning system in Kedah
-
-



LIMITATIONS

The study limits its scope to just two specific months, one representing the wettest period and the other the driest. While this approach highlights extreme weather conditions, it may not provide a comprehensive understanding of the climate or rainfall patterns throughout the entire year. The findings may not be fully applicable to other months, seasons, or regions with differing weather patterns.

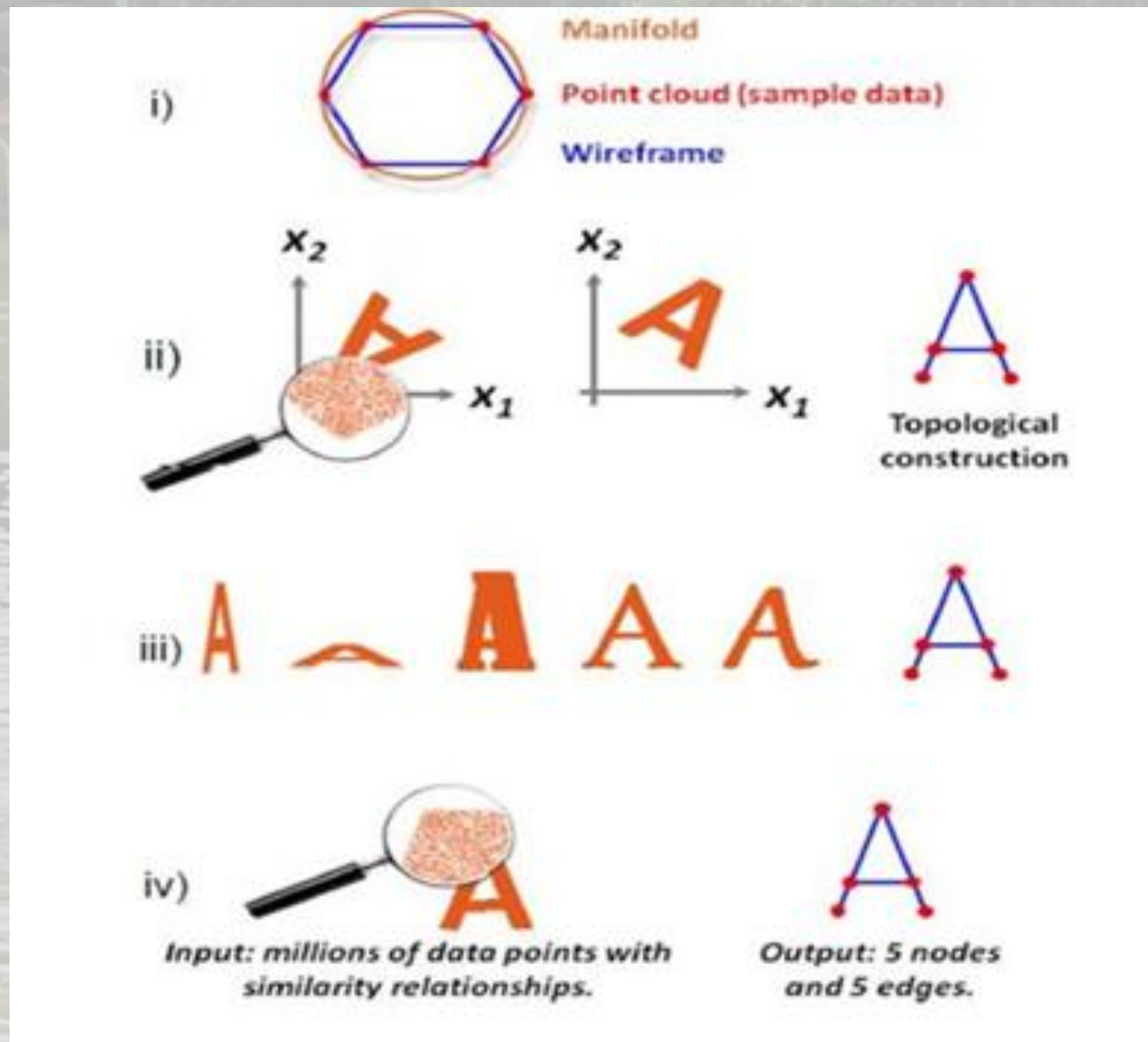
The study relies exclusively on the TDA (Topological Data Analysis) method for predicting future floods. While this method has its merits, it overlooks the potential benefits of incorporating other data analysis or modelling techniques, which could provide a more accurate, holistic view of flood predictions. Other factors, such as environmental or socio-economic data, are not considered in this analysis.

The entire study is carried out using a single coding software, which may limit the flexibility and scalability of the methods used. Different coding platforms may offer unique features, libraries, or tools that could enhance the analysis and results. Using only one software restricts the possibility of experimenting with other frameworks that might yield more robust or optimized outcomes.

LITERATURE REVIEW: TDA

Key Properties of TDA:

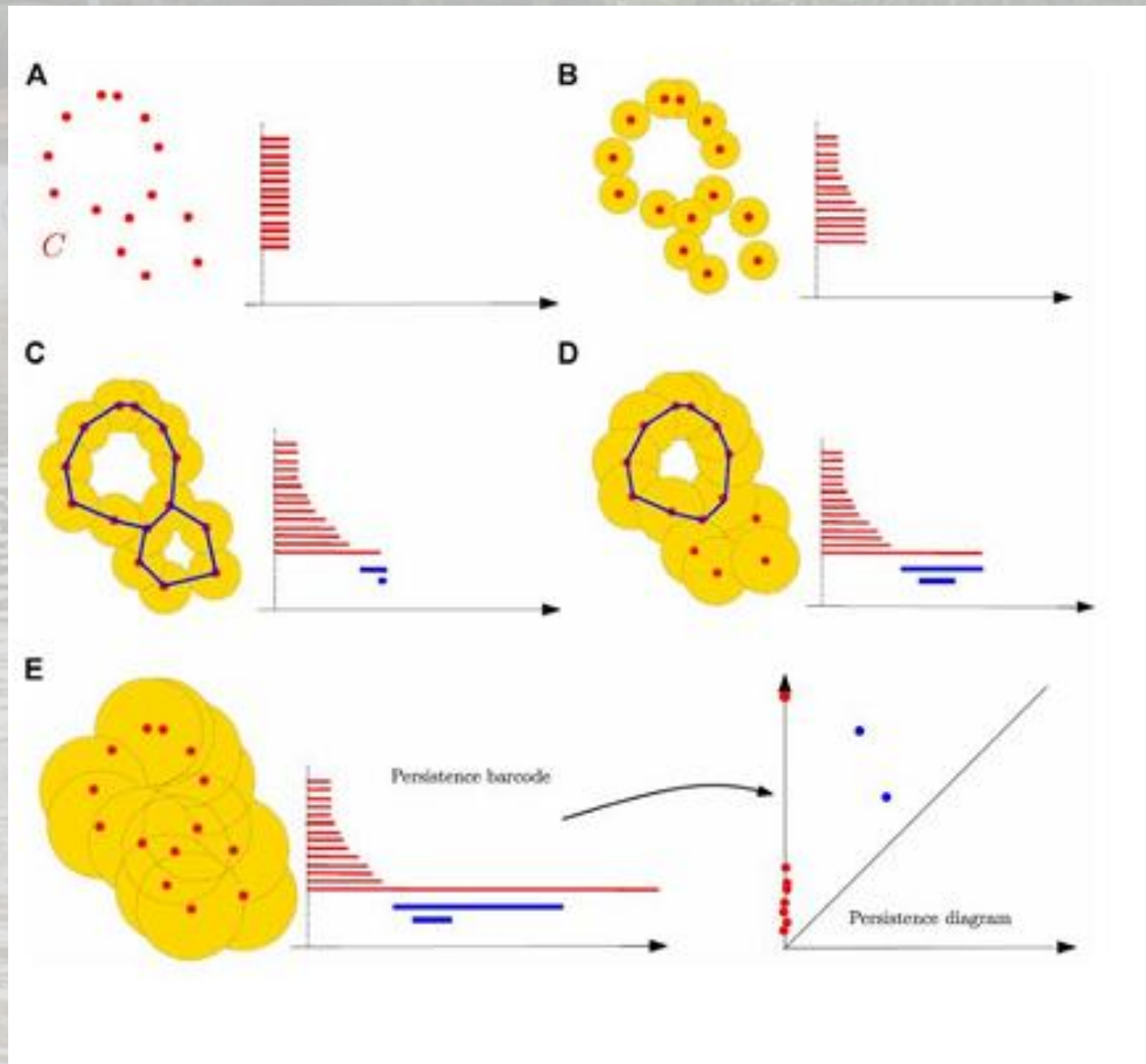
1. Invariance under rotation/coordinate changes: Topological measures remain the same despite changes in orientation or position.
2. Invariance under smooth deformation: Features persist through stretching or bending, capturing essential shape properties like loops and holes.
3. Compressed data representation: Simplifies data by focusing on persistent key features, enabling efficient analysis and visualization.



Vital idea of TDA and three important properties of TDA

(Source: Ohanuba et al., 2021)

LITERATURE REVIEW: PD



Point cloud of family of persistence
(Source: Chazal and Michel, 2021)

- Persistent Diagram (PD) represents birth and death of topological features across a filtering space. PD creates a family of persistence diagrams, showing how topology evolves.
- When the radius centered on data points increases, the structure of the space grows and the points merge into components, loops are formed and the space eventually fills in.
- These transitions are the visual representation of persistence barcode and diagram.

LITERATURE REVIEW

Study about feature representation analysis using TDA and coding of TDA

Author(s) Year(s)	Sample Size			Application			Remarks	Interpolation Method
	Small	Medium	Big data	K-Mean	TDA	Clustering		
Chazal and Michel (2021)	/				/		GUDHI library, Python, C++	
Khan et al., (2021)			/			/	IoT	/
Kim and Mémoli (2021)			/		/		Mathematical framework	
Skaf and Laubenbacher (2021)			/		/		TDA tools	
Ohanuba et al., (2021)		/			/	/	R software, Python	/

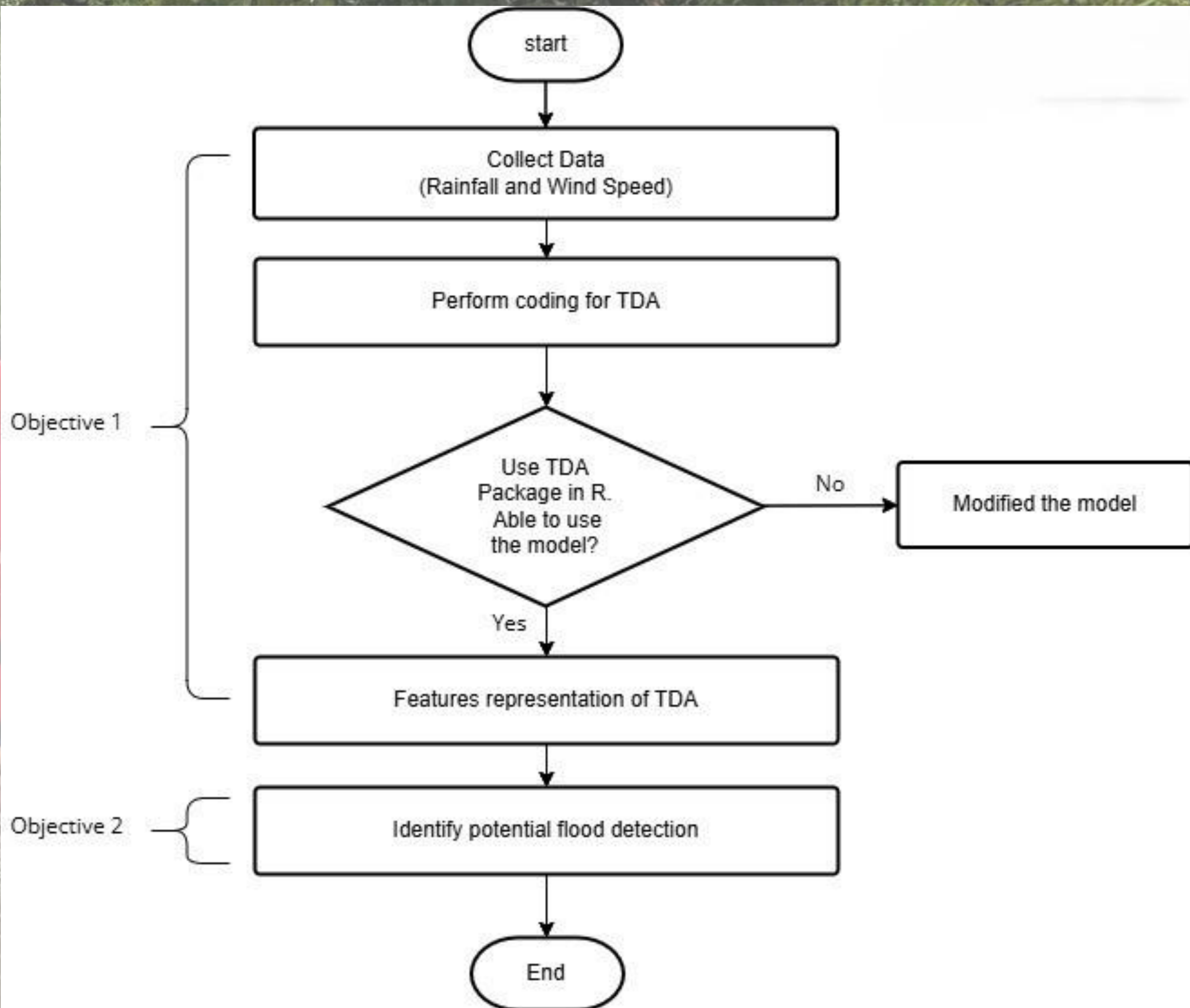
Author(s) Year(s)	Sample Size			Application			Remarks	Interpolation Method
	Small	Medium	Big data	K-Mean	TDA	Clustering		
Syed Musa et al., (2021)		/			/		R-package ‘TDA’	/
Ohanuba et al., (2022)			/	/	/	/	Python, R Software	/
Ren et al., (2022)			/		/		Mathematical framework	
Yan et al., (2022)	/				/		Python, SWMM	
Veres et al., (2023)			/		/		Python	/

LITERATURE REVIEW

Study about flood modelling

Author(s) Year(s)	Sample Size			Application		
	Small	Medium	Big Data	Sensor	Components	Others
Zhang et al., (2022)			/	Remote Sensing		
Veres et al., (2023)		/			Medical (Cardiology)	
Ibrahim et al., (2023)	/					Field-based Survey
Tanim et al., (2022)			/	Remote Sensing		ML
Refice et al., (2022)			/	Remote Sensing		
Patel et al., (2022)	/					IoT
Yadav et al., (2022)			/	Remote Sensing		Deep Learning
Farahmand et al., (2021)		/				Flood Control Infrastructure
Mokkenstorm et al., (2021)		/		Remote Sensing		
Ishiwatari et al., (2023)			/			Multiple sources in DRR contexts

Methodology



OPTIMAL DATA

	A	B	C	D	E
1	1.5	1.7	2		
2	1.5	1.5	1.7		
3	1.9	1.5	1.5		
4	1.9	1.9	1.5		
5	1.7	1.9	1.9		
6	1.5	1.7	1.9		
7	1.8	1.5	1.7		
8	1.6	1.8	1.5		
9	1.9	1.6	1.8		
10	1.9	1.9	1.6		
11	1.8	1.9	1.9		
12	2.2	1.8	1.9		
13	2.2	2.2	1.8		
14	1.8	2.2	2.2		
15	1.9	1.8	2.2		
16	0	0	0		
17	0	0	0		
18	0	0	0		
19	0	0	0		
20	0	0	0		
21	0	0	0		
22	0	0	0		
23	0	0	0		
24	0	0	0		
25	0	0	0		
26	0	0	0		
27	0	0	0		
28	0	0	0		

Optimal data of
February 2014

	A	B	C	D	E
1	86.7	84.8	83.5		
2	92.5	86.7	84.8		
3	95.7	92.5	86.7		
4	83.9	95.7	92.5		
5	87.3	83.9	95.7		
6	81.9	87.3	83.9		
7	79.2	81.9	87.3		
8	83.6	79.2	81.9		
9	87.5	83.6	79.2		
10	89.3	87.5	83.6		
11	87.8	89.3	87.5		
12	85.4	87.8	89.3		
13	88.7	85.4	87.8		
14	84.4	88.7	85.4		
15	84	84.4	88.7		
16	38.4	7.4	3.4		
17	1.2	38.4	7.4		
18	9.2	1.2	38.4		
19	1.6	9.2	1.2		
20	10.8	1.6	9.2		
21	2.4	10.8	1.6		
22	20.6	2.4	10.8		
23	54.6	20.6	2.4		
24	88.6	54.6	20.6		
25	0.4	88.6	54.6		
26	7	0.4	88.6		
27	3	7	0.4		
28	0.4	3	7		
29	0	0.4	3		
30	0	0	0.4		
31	0	0	0		

Optimal data of
October 2014

Source :

Jabatan Meteorologi Malaysia,
(Records of Meteorological Data)

Station : Alor Setar

Lat. : 6° 12' N

Long. : 100° 24' E

Ht. above M.S.L. : 3.9 m

MATHEMATICAL FORMULATION

```
FUNCTION CalculatePersistentDiagramFeatures(q)
  // q = {(b_i, d_i)} in dimension D
  n = LENGTH(q)

  // Initialize features
  max_D = 0
  sum_D = 0
  avg_D = 0
  max_D_death = 0

  // First feature: Maximum holes lifetime
  FOR i = 1 TO n DO
    lifetime = q[i].d - q[i].b
    IF lifetime > max_D THEN
      max_D = lifetime

  // Second feature: Sum of holes lifetime
  FOR i = 1 TO n DO
    sum_D = sum_D + (q[i].d - q[i].b)

  // Third feature: Average lifetime of holes
  IF n > 0 THEN
    avg_D = sum_D / n

  // Fourth feature: Maximum death point of holes
  FOR i = 1 TO n DO
    IF q[i].d > max_D_death THEN
      max_D_death = q[i].d

  RETURN {max_D, sum_D, avg_D, max_D_death}
END FUNCTION
```




R Programming

R is a programming language for statistical computing and data visualization. It has been widely adopted in the fields of data mining, bioinformatics, data analysis, and data science.

Key features include powerful data handling, advanced statistical and graphical capabilities, and a vibrant community-driven extension system.

BUILT-IN FUNCTION

Function	Description
DiagLim	This parameter sets the maximum filtration value for the calculation of persistent homology
maxdimension	Specifies the maximum dimension for features.
ripsDiag	The main function for computing persistent homology based on the Rips complex. It takes data and computes the persistence diagram or barcode for the given data set.
plotBarcode	A visualization function for plotting the barcode diagram that represents the persistence of features at various scales.
plotDiagram	A function for plotting the persistence diagram, which shows the birth and death of features in the dataset.

CODING

```
1 # call library
2 library(TDA)
3
4 #read data
5 feb2014AS <- read.csv("C:/Users/AIN/Desktop/FYP/Data/feb2014AS.csv", header = FALSE)
6 View(feb2014AS)
7
8 # set filter value and maximum dimension
9 DiagLim <- 700
10 maxdimension <- 1
11
12 # Apply persistent homology using Rips Diagram
13 feb2014 <- ripsDiag(feb2014AS,maxdimension,DiagLim,printProgress = TRUE)
14
15 # Plot barcode and persistent diagram
16 plot(feb2014[["diagram"]],barcode=TRUE,main="Barcode Feb 2014")
17 plot(feb2014[["diagram"]],main="PD Feb 2014")
18
19 # save diagram as matrix
20 matrixfeb2014 <- as.data.frame.matrix(feb2014[["diagram"]])
21 matrixfeb2014
22
23 # separate matrixfeb2014 for dimension 0 only
24 a <- matrixfeb2014$dimension == 0
25 feb2014dim0 <- matrixfeb2014[a,]
26
27 # calculate lifetime of all feature
28 feb2014dim0$difference <- feb2014dim0$Death-feb2014dim0$Birth
29 feb2014dim0
30
31 sum(feb2014dim0 $difference)-DiagLim
32
33 avg <- (sum(feb2014dim0 $difference)-DiagLim)/(nrow(feb2014dim0)-1)
34 avg
35
36
37 # calculate topology for dimension 1
38 b <- matrixfeb2014$dimension==1
39 feb2014dim1 <- matrixfeb2014[b,]
40
41 #calculate lifetime for all features
42 feb2014dim1$difference <- feb2014dim1$Death-feb2014dim1$Birth
43 feb2014dim1
44
45 #calculate topological features
46 sum(feb2014dim1$difference)
47
48 avg <- sum(feb2014dim1$difference)/nrow(feb2014dim1)
49 avg
50
51 max(feb2014dim1$difference)
52 max(feb2014dim1$Death)
53
```

```
1 # call library
2 library(TDA)
3
4 #read data
5 oct2014AS <- read.csv("C:/Users/AIN/Desktop/FYP/Data/oct2014AS.csv", header = FALSE)
6 View(oct2014AS)
7
8 # set filter value and maximum dimension
9 DiagLim <- 700
10 maxdimension <- 1
11
12 # Apply persistent homology using Rips Diagram
13 oct2014 <- ripsDiag(oct2014AS,maxdimension,DiagLim,printProgress = TRUE)
14
15 # Plot barcode and persistent diagram
16 plot(oct2014[["diagram"]],barcode=TRUE,main="Barcode Oct 2014")
17 plot(oct2014[["diagram"]],main="PD Oct 2014")
18
19 # save diagram as matrix
20 matrixoct2014 = as.data.frame.matrix(oct2014[["diagram"]])
21
22 # separate matrixoct2014 for dimension 0 only
23 a = matrixoct2014$dimension == 0
24 oct2014dim0 = matrixoct2014[a,]
25
26 # calculate lifetime of all feature
27 oct2014dim0$difference = oct2014dim0$Death-oct2014dim0$Birth
28 oct2014dim0
29
30 sum(oct2014dim0 $difference)-DiagLim
31
32 avg=(sum(oct2014dim0 $difference)-DiagLim)/(nrow(oct2014dim0)-1)
33 avg
34
35 # calculate topology for dimension 1
36 b=matrixoct2014$dimension==1
37 oct2014dim1=matrixoct2014[b,]
38
39 #calculate lifetime for all features
40 oct2014dim1$difference=oct2014dim1$Death-oct2014dim1$Birth
41 oct2014dim1
42
43 #calculate topological features
44 sum(oct2014dim1$difference)
45
46 avg=sum(oct2014dim1$difference)/nrow(oct2014dim1)
47 avg
48
49 max(oct2014dim1$difference)
50
51 max(oct2014dim1$Death)
```

Code for February and October 2014

RESULTS: OPTIMIZED DATA

	A	B	C	D	E	F
1	Stnno	Year	Month	Day	Rainfall (mm)	Mean Wind (m/s)
2	48603	2014	2	1	0	2
3	48603	2014	2	2	0	1.7
4	48603	2014	2	3	0	1.5
5	48603	2014	2	4	0	1.5
6	48603	2014	2	5	0	1.9
7	48603	2014	2	6	0	1.9
8	48603	2014	2	7	0	1.7
9	48603	2014	2	8	0	1.5
10	48603	2014	2	9	0	1.8
11	48603	2014	2	10	0	1.6
12	48603	2014	2	11	0	1.9
13	48603	2014	2	12	0	1.9
14	48603	2014	2	13	0	1.8
15	48603	2014	2	14	0	2.2
16	48603	2014	2	15	0	2.2
17	48603	2014	2	16	0	1.8
18	48603	2014	2	17	0	1.9
19	48603	2014	2	18	0	1.9
20	48603	2014	2	19	0	1.5
21	48603	2014	2	20	0	2.2
22	48603	2014	2	21	0	2.4
23	48603	2014	2	22	0	2.9
24	48603	2014	2	23	0	2.5
25	48603	2014	2	24	0	2.2
26	48603	2014	2	25	0	2.5
27	48603	2014	2	26	0	2.3
28	48603	2014	2	27	0	2.7
29	48603	2014	2	28	0	2.2
30						
< > FEB2014 +						

	A	B	C	D	E	F
1	Stnno	Year	Month	Day	Rainfall (mm)	Mean Wind (m/s)
2	48603	2014	10	1	0	83.5
3	48603	2014	10	2	2.8	84.8
4	48603	2014	10	3	4.8	86.7
5	48603	2014	10	4	12.4	92.5
6	48603	2014	10	5	43.8	95.7
7	48603	2014	10	6	6.8	83.9
8	48603	2014	10	7	0	87.3
9	48603	2014	10	8	0	81.9
10	48603	2014	10	9	0	79.2
11	48603	2014	10	10	0	83.6
12	48603	2014	10	11	0.8	87.5
13	48603	2014	10	12	13.2	89.3
14	48603	2014	10	13	3.6	87.8
15	48603	2014	10	14	12	85.4
16	48603	2014	10	15	2.8	88.7
17	48603	2014	10	16	3.4	84.4
18	48603	2014	10	17	7.4	84
19	48603	2014	10	18	38.4	87.8
20	48603	2014	10	19	1.2	86.3
21	48603	2014	10	20	9.2	85.5
22	48603	2014	10	21	1.6	87.9
23	48603	2014	10	22	10.8	90.5
24	48603	2014	10	23	2.4	91.2
25	48603	2014	10	24	20.6	84.9
26	48603	2014	10	25	54.6	84.4
27	48603	2014	10	26	88.6	83.8
28	48603	2014	10	27	0.4	86.4
29	48603	2014	10	28	7	85.3
30	48603	2014	10	29	3	84
31	48603	2014	10	30	0.4	85.4
32	48603	2014	10	31	0	85.2
33						
< > OCT2014 +						

Typical rainfall intensities (mm):

- No Rain : 0
- Light Rain : 0 < Rainfall < 10
- Heavy Rain : > 10

Optimized data for February and October 2014

RESULTS: FEATURE REPRESENTATION (FEBRUARY 2014)

Topology features (Dimension 0)

	dimension	Birth	Death	difference
1	0	0	700.0000000	700.0000000
2	0	0	2.7184554	2.7184554
3	0	0	0.4123106	0.4123106
4	0	0	0.4123106	0.4123106
5	0	0	0.3741657	0.3741657
6	0	0	0.3316625	0.3316625
7	0	0	0.3162278	0.3162278
8	0	0	0.3162278	0.3162278
9	0	0	0.3000000	0.3000000
10	0	0	0.2828427	0.2828427
11	0	0	0.2828427	0.2828427
12	0	0	0.2236068	0.2236068
13	0	0	0.1732051	0.1732051
14	0	0	0.1000000	0.1000000
15	0	0	0.1000000	0.1000000
16	0	0	0.1000000	0.1000000

Summary statistics (Dimension 0)

Sum of all lifetimes (sum _D)	Average of lifetime of all holes (avg _D)
6.443858	0.4295905

Topology features (Dimension 1)

	dimension	Birth	Death	difference
17	1	0.3316625	0.4242641	0.09260159
18	1	0.3741657	0.4242641	0.05009833
19	1	0.4000000	0.4123106	0.01231056

Summary statistics (Dimension 1)

Max. holes lifetimes (max _D)	Max. death point of holes (max _d)	Sum of all lifetimes (sum _D)	Average of lifetime of all holes (avg _D)
0.1550105	0.4242641	0.09260159	0.05167016

RESULTS: FEATURE REPRESENTATION (OCTOBER 2014)

Topology features (Dimension 0)

	dimension	Birth	Death	difference
1	0	0	700.000000	700.000000
2	0	0	68.909506	68.909506
3	0	0	65.392431	65.392431
4	0	0	51.412450	51.412450
5	0	0	50.254552	50.254552
6	0	0	29.246538	29.246538
7	0	0	28.228355	28.228355
8	0	0	20.920803	20.920803
9	0	0	19.914819	19.914819
10	0	0	10.721940	10.721940
11	0	0	9.961928	9.961928
12	0	0	8.654479	8.654479
13	0	0	7.731753	7.731753
14	0	0	7.615773	7.615773
15	0	0	6.891299	6.891299
16	0	0	6.469930	6.469930
17	0	0	5.590170	5.590170
18	0	0	4.787484	4.787484
19	0	0	4.549725	4.549725
20	0	0	4.535416	4.535416
21	0	0	4.041039	4.041039
22	0	0	4.013726	4.013726
23	0	0	3.749667	3.749667
24	0	0	3.725587	3.725587
25	0	0	3.509986	3.509986
26	0	0	3.354102	3.354102
27	0	0	3.234192	3.234192
28	0	0	2.727636	2.727636
29	0	0	2.630589	2.630589
30	0	0	1.833030	1.833030
31	0	0	0.400000	0.400000

Summary statistics (Dimension 0)

Sum of all lifetimes (sum_D)	Average of lifetime of all holes (avg_D)
445.0089	14.83363

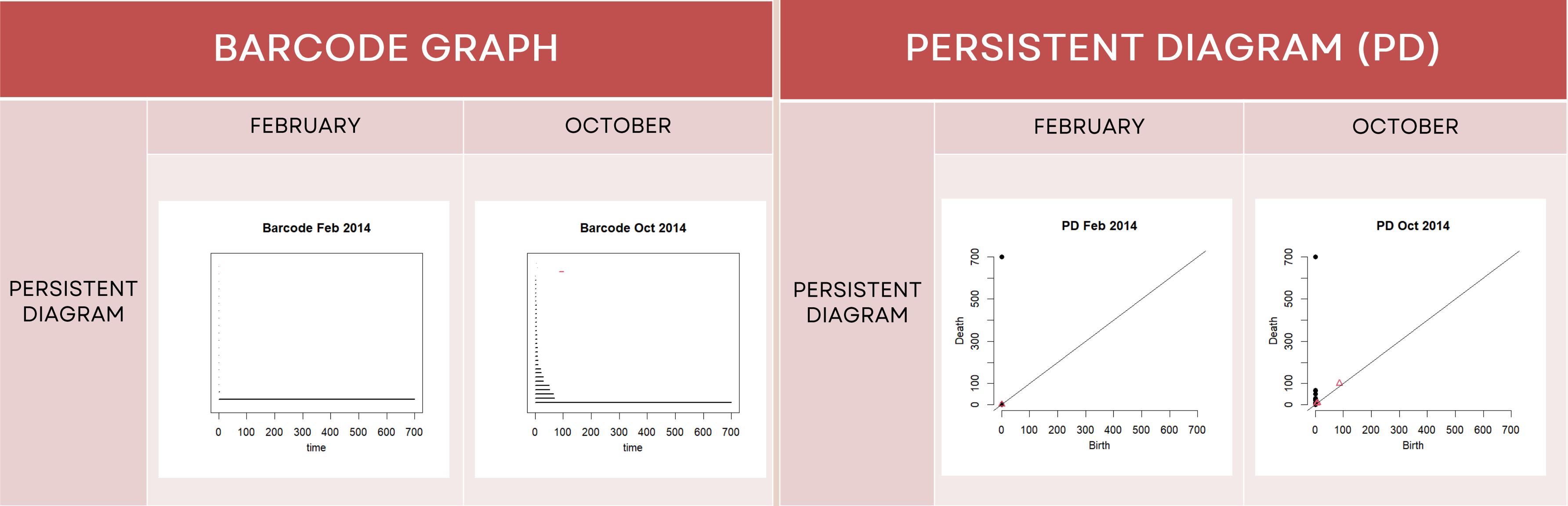
Topology features (Dimension 1)

	dimension	Birth	Death	difference
32	1	85.578151	100.455164	14.8770127
33	1	7.470609	8.514693	1.0440841
34	1	4.910193	5.356305	0.4461112
>				

Summary statistics (Dimension 1)

Max. holes lifetimes (max_D)	Max. death point of holes (max_d)	Sum of all lifetimes (sum_D)	Average of lifetime of all holes (avg_D)
16.36721	100.4552	14.87701	5.455736

RESULTS: PERSISTENT HOMOLOGY



Barcode for no flooding and flooding months in Alor Setar

PD for no flooding and flooding months in Alor Setar

CONCLUSION

- Persistent Homology (PH) analyzes environmental time-series data to identify key flood features by measuring their persistence, using barcode plots and Persistent Diagrams (PD) to visualize connected components and holes.
- PH improves flood detection by capturing patterns that traditional methods miss, helping identify flood-prone areas and changes in water flow.
- TDA with PH provides a strong foundation for early warning systems and accurate flood forecasts, supporting better flood mitigation strategies.

FUTURE WORK

- Apply TDA-based flood prediction methods to other countries with severe weather challenges.
- Combine TDA with other methods to improve model flexibility and accuracy.
- Enhance TDA's statistical reliability by using stronger statistical approaches.



Thank you